

some 16 percent of grown men from most native communities lying between Cuzco and Potosí were ordered to move to the mines and stay there for a year. The total number was some 13,500; but each man had to work only one week in three, so that some 4,500, or slightly over 5 percent, were actively laboring in mining or refining at any moment.

In their initial forms, these drafts may actually have spread the labor burden more evenly among the native population than the *encomienda* had done, since some *encomenderos* worked their people pitilessly, and others lightly. Another gain over earlier "personal service" was that draftees received a statutory wage. It was probably not a living wage for a family; but in the *encomienda*, with the rarest exceptions, no wage had been paid – the intended reward of instruction in Christianity and European ways was intangible, and rarely delivered. In principle, also, draft labor should have given Indians some protection from abusive employers, since the state, represented by an allocating official, stood between employer and worker. But the benefit in practice may have been small. From the administration's standpoint, drafts offered two economic gains. One was that the shrinking labor supply could be channeled to what was considered essential production, defined as "public works." These included, certainly, the building of roads, bridges, drainage works, churches, and other public edifices; but, more crucial to both state and individuals, they also included agriculture and mining. Second, drafts broke the close control over labor that *encomenderos* had exercised; now workers could be given to incoming Spaniards who might well be more entrepreneurial than aging conquistadores and their socially aspiring offspring.

In the areas of intense economic activity by the Spanish, *mita* and *repartimiento* proved, though, inadequate almost as soon as put in place. In New Spain, severe epidemics, possibly of typhus, shrank the remaining Indians' numbers drastically in the late 1570s. In the central Andes also, the native population continued to fall, while the demand for workers at Potosí and other silver mines, at Huancavelica (the source of the indispensable mercury), and generally for farming (to feed the booming mining towns), leapt upward. Constant immigration from Spain created further imbalance between supply and demand in labor. One outcome was that the draft quotas increased in the closing years of the century. By 1600 some draftees were working every other year at Potosí, instead of one in seven. A parallel result, clear in its economic logic, was that Spaniards needing workers hired them individually wherever they could find them. Since almost all adult native men were liable for drafting, such private hiring cut into the numbers available for forced work – from which Indians were ever more anxious to escape, of course, as the "take" increased. By about 1600, then, working for wages was well ensconced in regions and activities which had a high demand for labor. The numbers are clearest in silver-mining. In New Spain the total workforce was counted c.1597 at 9,143, of whom 6,261 were waged laborers (68.5 percent), 1,619 *repartimiento* men (17.7 percent), and 1,263 black slaves (13.8 percent). In Potosí, a few years later, the total daily workforce was about 9,900, of whom some 55 percent were waged. If other mines by then active in the Potosí district are

taken into account, received no draft; black slaves; the chill to make in the central beginning of the

Hiring Indians colonialists. Some artisanry received thin and scattered rose in fact for price of labor as, under most their employees competed for their advances on the workplace as the

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taken into account, the proportion of wage workers rises, since those mines received no draft. Miners and refiners in the central Andes used hardly any black slaves; they reportedly succumbed too soon to disease in the high-altitude chill to make buying them a good investment. And in neither New Spain nor in the central Andes were any *encomienda* Indians working in mining by the beginning of the seventeenth century.

Hiring Indians for wages spread across productive activities controlled by colonials. Some native workers in textile shops, some on estates, and some in artisanry received negotiated cash sums for their labors. The evidence, though thin and scattered, is also that such workers' wages rose in the late 1500s, and rose in fact faster than prices, so that their real income grew.⁴⁵ The relative price of labor should certainly have increased as workers became scarcer, and as, under more or less permanent contract, they acquired skills valuable to their employers. So desirable did skilled workers become that employers competed for them, luring them away with offers of higher wages. In some cases advances on wages were given not so much to tie a worker through debt to a workplace as to attract him to it in the first place.

Thus, in the regions most heavily developed economically by colonists (which corresponded generally with the areas of prior high native culture, but expanded now by silver discoveries), employment of Indians moved in the sixteenth century from the retainer-like relationship of the *encomienda*, to the state-run (but salaried) draft of *mita* or *repartimiento*, to the private and individual contracts of wage labor. A variety of causes drove this progression: the crown's reliance on, then rejection of, *encomienda* as a method of settlement; rising immigration from Spain; and, most powerful of all, the effects of native depopulation in, first, shrinking the pool of available workers, and, second, forcing the colonists to start organizing essential production themselves.

Even in the core colonial areas of New Spain and the central Andes, however, the coming of a new stage did not necessarily mean the disappearance of a previous one. *Mita* labor was crucial to Andean silver-mining to the end of colonial times, because it provided a cheap and reliable supplement to wage labor. Similarly, in New Spain, although the then viceroy ended drafts for agriculture in 1632 because they had become unnecessary, *repartimiento* for silver-mining remained active, at least for mines in the center and south.

And in peripheral regions, where whites and their enterprises were fewer and less dominant, use of native labor took a variety of courses. A common feature was survival of the *encomienda* as a source of labor, despite the decree halting personal service in 1549.⁴⁶ In outlying parts of New Spain such as New Mexico and Yucatan, as well as in remoter regions of South America, such as Venezuela, New Granada, Paraguay, and Chile, it little mattered to the crown if the *encomienda* persisted in its initial form, since there was little to be lost there, politically or economically, to a well-rooted seigneurial elite in society. In fact, it suited the crown's purpose that the *encomienda* should continue to play its anchoring role in those more or less marginal places, holding settlers usefully in place on the land. It was typical of the peripheries, also, that unless some new economic resource or new export opportunity appeared, competition

for labor was slight; hence the economic need to move from *encomienda* to draft was small, or arrived much later than in Peru and New Spain. Again, the marginal areas had had small, weak, or no state structures before the conquest, and little in the way of state-organized labor; so there was no precedent in them to help the Spanish to set up their own drafts.

A blend of *encomienda* and *repartimiento*, whereby *encomenderos* were the main recipients of draft labor, emerged in Paraguay and New Granada. In Chile, by contrast, while *encomienda* continued to provide laborers until the late eighteenth century, drafts were not used. Colonists not holding *encomiendas* rented workers from those who did, in a practice suggesting the reduction of *encomienda* Indians almost to chattel status. It was, in fact, precisely in Chile that outright enslavement of Indians persisted most strongly. After the great native revolt in the south beginning in 1598, slaving was specifically permitted; and until abolition in 1674, Indians taken in war were legally put to work as slaves. Indian slavery had generally been ended in the empire c.1550 because of its blatant contradiction of the principle, by then well established, that Indians were intrinsically free, and also, perhaps, because slavery was a labor system afflicted with the same rigidities as *encomienda*, when what was needed was the flexibility in allocating workers that the draft provided. But an old policy persisted that Indians who rebelled against Spanish rule could legitimately be enslaved. This meant continued Indian slaving not only in the southern extreme of the empire but in the far north as well, when Comanches and Apaches raided New Mexico. Calling this "rebellion" bordered on sophistry, since those peoples had never been under Spanish control. But it was a convenient fiction, for it provided a steady, if small, supply of slaves to northern New Spain on into the seventeenth century while also helping to sustain settlement in the distant north, since colonists profited from selling the slaves they took.

BLACK SLAVERY

In aggregate, though, Indians were far outnumbered as slaves by blacks in the sixteenth century, as slave imports from Africa rose with the passing years. And those blacks were an increasingly large supplement to native labor in its various forms, as the indigenous numbers fell. Their participation in higher-level occupations, such as management and artisanry, was important as a social as well as an economic phenomenon. But where they became all but essential to the colonial economy was in providing labor that native people could not, for whatever reason, supply. The extreme example is agricultural work, particularly in sugar, on the large Caribbean islands, where literally no indigenous people remained after the mid sixteenth century. Sugar cultivation was the work almost entirely of black slaves everywhere, for at least two reasons. The warmer, lowland areas in which the cane thrived, such as the Mexican and Peruvian coasts, had suffered more severe losses of Indians than the highlands; and, at least in the Spaniards' stereotypical view, Africans withstood hot, heavy

work better than of prior native to black slaves accumulate in Granada, the of blacks. So Venezuela.⁴⁷

TRADE IN

The products of sixteenth-century farming their. But, when ordinary men might find that would be further profound (besides technological) immense growth unknown in the Honduras Central America up and down the trade, grew vast with Central America the Río de la Plata oxen, horses, and change, offering it, of course that trade was once validated over distance goods encouraged be grown or comparative advantage and climate for by cart to central there. Southern there could be New Spain), and fully by the late freight cost. La across the empire northern New Spain the sugar areas

work better than Indians. That same opinion, added to the generally low level of prior native political, and hence labor, organization in tropical areas, led also to black slaves' often being used to pan gold, since alluvial gold tended to accumulate in the lower reaches of rivers. Hence gold extraction in New Granada, the metal's main Spanish American source, became largely the work of blacks. So was the cultivation of cacao in southern lowland Quito and in Venezuela.⁴⁷

TRADE IN THE COLONIES

The products of the variety of workers and of working arrangements found in sixteenth-century Spanish America reached a wide range of consumers. Indians farming their communities' lands produced largely for their own subsistence. But, when ordered by Spanish officials to work in *repartimiento*, those same men might find themselves on a colonist's estate plowing land to grow wheat that would be sent to a market a few, or a few hundred, kilometers away. One further profound difference between economic practice in 1500 and, say, 1600 (besides technology, forms of land tenure, labor systems, and the like) was an immense growth in exchange over distance. Such exchange had not been unknown in native times, of course. Columbus met coastal trading canoes off the Honduras coast in 1502; Aztec *pochteca* brought the products of northern Central America to Tenochtitlan; the Inca state directed movement of goods up and down the Andes. But the distances, as well as the size and variety of trade, grew vastly once the Spanish came. Peru had never before dealt directly with Central America and Mexico; nor the Andean province of Charcas with the Río de la Plata; nor the Greater Antilles with Panama. Clearly ships, carts, oxen, horses, and mules were the key to this immense expansion of interchange, offering swift and relatively cheap carriage of large volumes. Also propelling it, of course, was the belief, grounded in much European experience, that trade was the surest source of profit. Spanish American colonial experience validated that belief from the earliest days until the last. Then again, trade over distance grew and thrived precisely because the possibility of moving goods encouraged settlers to look for places where this or that item could easily be grown or made. Thus they found and developed large areas enjoying comparative advantages for some particular purpose. The Bajío proved ideal in soil and climate for wheat and maize, and the grain could be moved cheaply enough by cart to central Mexico or the northern mines to allow for profitable sale there. Southern Peru offered excellent conditions for grapes; wine produced there could be cheaply sent northward by ship to Lima and beyond (even to New Spain), and by mule to the Andean mines, where it competed successfully by the late sixteenth century with Spanish wine that carried a still higher freight cost. Large and specialized zones of production were in clear evidence across the empire before 1600: the two just mentioned; the cattle lands of northern New Spain; the sheep lands of Puebla, and of Quito in South America; the sugar areas of the Antilles, southern New Spain, and northern Peru; and

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the developing mule and cattle lands of Tucumán in present northern Argentina. In an impressive realization of the potential of economic geography, production fed trade and trade stimulated production in these naturally favored areas.

Most of the exchange of these agricultural products, and manufactures made from them, was within the colonies, either within single *audiencia* territories or between them. The most powerful magnets of trade were the administrative cities and the mining towns – especially the latter, because they both tended to have large populations and were the source of the most desirable commodity exchangeable for goods: silver (whether in unminted or coined form). By one historian's rough estimate, a little over half the silver refined in Spanish America went to buy goods produced in the colonies; and these regional purchases by mining centers amounted to 60 or 70 percent of the value of transoceanic trade to and from the mining areas.⁴⁸ If these calculations are right (and much descriptive evidence argues for them), bullion produced in America did not, as often assumed, flood directly out to foreign destinations in Europe and the Orient, but instead passed through and energized internal trade circuits before it left America. Moreover, far from all of it did leave, though the fraction remaining undoubtedly varied with time.

OCEANIC TRADE

What sixteenth-century Europeans saw, however, and East Asians soon after them also, as they looked towards Spanish America, was not a developing structure of internal exchange, but shiploads of gold and silver crossing the oceans towards them. The transatlantic trading system of Spain, and then the trans-Pacific commerce, have likewise long captured the attention of historians; and with good reason, since these were maritime operations to rival in scale and pomp anything before or since, and economic movements that produced irreversible effects on Europe and the wider world.

The small groups of ships used by the explorers gradually expanded with discoveries and conquests into flotillas, then into fleets, and finally, in the second half of the century, into great armadas carrying people and European cloth, tools, wine, foods, and other exports to America, and silver, dyes, leather, sugar, and medicinal plants back to Spain. In 1526 sailings of single merchant ships to and from the Indies were forbidden. Convoys, for safety from enemy attack and for best use of the limited number of pilots capable of mathematically based navigation, were the rule from then on. By the best estimate, at least 85 percent of Spanish transatlantic sailing from 1500 to 1650 was done in convoy.⁴⁹ Security increased from c.1540 with the entry into service of galleons, ships initially of perhaps 250 tons (though they reached 800 by the end of the century) in which some of the fine line of Mediterranean galleys was combined with the seagoing ability of Atlantic trading vessels to produce a relatively fast and handy fighting ship that could also carry cargo.

By the mid 1560s the classic Spanish American "fleet system" was in place, undertaking the Carrera de Indias, the "Indies run," each year. Two fleets were

the backbone of the the *flota* (simply "fleet" (galleons), whose destination was the Indies. Both fleets might number as many as 100 ships. The fleets included galleons, ships that carried silver. The isthmus-based fleets included a fraction of galleons, that South American ports.

The standard rhythm should leave Spain in the Canaries, following a route through the Lesser Antilles or the Gulf of Mexico, the Greater Antilles and back to Spain. Goods; winter in the Indies; spring; and make the Azores, and again follow the same route. If they stuck to the early autumn hurricane season, winter storms further west, Bermuda suggests, leaving Spain, enemy return voyage, mere the rhythm of the C tioned as intended, goods, bullion, and, Atlantic in the sixteenth century.

If all went exactly around the departure not possibly be refitted. So shipping was necessary sustained. This meant Mexico spent only a *galeones* must have been ships and their cargo and small-group sailing.

It was also the priority of the Casa de C with the Indies that all Spanish contact powerful merchant especially those of the of Seville seized on that fleets should sail channel of trade with

the backbone of the communication structure between Spain and America: the *flota* (simply "fleet") sailing to and from New Spain, and the *galeones* (galleons), whose destination was the Caribbean side of the Isthmus of Panama. Both fleets might consist of several dozen ships, though as important as the number was the growing size of the vessels as the century wore on. And both fleets included galleons, for defense and for freight of the main return cargo, silver. The isthmus-bound *galeones* were so called because that fleet had a larger fraction of galleons, to bring back safely the larger fraction of American silver that South America produced.

The standard rhythm of the Carrera was, broadly, as follows: the fleets should leave Spain in the early to mid summer; cross the Atlantic via the Canaries, following Columbus's route; enter the Caribbean in late summer through the Lesser Antilles; proceed to their appointed destinations on the isthmus or the Gulf coast of Mexico (with some ships dropping away to the Greater Antilles and Venezuela); discharge cargoes and load homebound goods; winter in the Caribbean; combine at Havana in the late winter or early spring; and make the eastward crossing, via the Florida Strait and perhaps the Azores, and again following Columbus's pattern, in the spring and early summer. If they stuck to this timetable, the fleets avoided the late summer and early autumn hurricanes of the Caribbean and western Atlantic, as well as the winter storms further north. As the goodly number of wrecks off Florida and Bermuda suggests, however, the schedule was not easily followed. Delays in leaving Spain, enemy presence, slow arrival of cargoes (especially silver) for the return voyage, mere logistical complexity – all these could and did interrupt the rhythm of the Carrera. Nonetheless, broadly speaking the design functioned as intended, transporting hundreds of thousands of people, animals, goods, bullion, and, not least, information and administrative orders across the Atlantic in the sixteenth century.

If all went exactly as planned, the combined returning fleet reached Spain at around the departure date of the next sailing. The same ships, however, could not possibly be refitted and reloaded in time to sail again that same summer. So shipping was needed for four distinct fleets if the annual round were to be sustained. This meant much time spent lying in port. Ships in the *flota* to Mexico spent only a quarter of their time at sea,⁵⁰ and the proportion for the *galeones* must have been similar. The cost in idle capital locked in motionless ships and their cargoes was high. But that was the price of safety. Single-ship and small-group sailings were seen as simply too risky.

It was also the price of the monopoly structure of the trade. After the creation of the Casa de Contratación in Seville in 1503 to regulate communication with the Indies that city gradually became, in law and fact, the sole conduit for all Spanish contact with America. It was already a great trading center, with a powerful merchant guild, and branches of many European trading houses, especially those of Italy, the Genoese prominent among them. The merchants of Seville seized on the American trade as their own province. It suited them that fleets should sail only at long, predictable intervals, and that no other legal channel of trade with America should exist, for that enabled them to drive up

prices by withholding goods from the transatlantic market. It suited them also that there should be only a few legal points of entry to the Indies, for that also helped them to control supply. All goods entering New Spain had to be unloaded at Veracruz, the sole destination of the *flota*; and all goods destined for Peru and elsewhere on the west coast of South America had to go in the *galeones* to the isthmus, for freighting across to the City of Panama, and loading there into a legally regulated Pacific fleet. Crown and merchants were in full agreement in these matters. What concerned the government above all was that its silver – the royal fifth and other taxes – should not escape from its grasp, and that customs duties on the American trade should be collected. Both aims were more easily achieved if contact with the Indies were concentrated into a few sailings, and funneled through a minimum of access points. If the crown should by chance lean towards a loosening of the fleet system, it would increasingly find itself reminded, as the years advanced, that the merchants of Seville were a valuable source of credit to the state, and that a weakening of the Seville monopoly would benefit neither them nor the treasury.

The closed structure of the Carrera in fact served crown and merchants well in the sixteenth century, better than in any later period. The trade grew in volume and value almost without interruption from 1505 to 1610 (c.23,000 tons of outward- and homeward-bound shipping from 1505 to 1509, and c.228,000 tons from 1605 to 1609).⁵¹ The causes of the growth were several: growing demand in America for Spanish and other European goods as a result of emigration and natural increase of the colonizing population; rising bullion production, giving colonists the means of paying for imports; colonists' growing engagement in economic activities as time passed, creating a demand for European tools and materials, such as metals; increasing and spreading wealth among settlers, spent partly on European luxury goods; and underlying the whole transatlantic exchange, of course, Spain's domination of the sea. Pirates and semi-official raiders might slip into the Caribbean, as Frenchmen and Englishmen did before 1600, and snatch substantial prizes to the Spaniards' irritation and alarm. But no nation could yet seriously threaten the great trading fleets at sea. The failure of the Armada sent against England in 1588 was a warning of things to come; but the lesson was not clear until the new century was well begun.

Possibly the most serious threat to the transatlantic trade in the late sixteenth century was not a military one from Spain's growing number of European enemies, but a commercial one from colonials' enterprises in the Pacific. After the discovery of a practicable route between New Spain and the Philippines in the mid 1560s, entrepreneurial traders in Mexico City soon decided to try their fortune in the Orient. The mercantile gateway to the east was Manila, founded on the island of Luzon in 1571 as the Spanish capital in the Philippines. Oriental goods reached Mexico from Manila in 1573, carried by the first of the Manila galleons. Those were great sturdy vessels that were soon being built in the Philippines from local teak. What above all energized this trans-Pacific trade was the relative shortage, and hence high valuation, of silver in the East. The Chinese valuation of silver in terms of gold in 1572, for

instance, was 1 A broad range buying them with Peru, where Portugal boom of all tin soon found the middle men in shipped into the Asia. Some of it at Macao that the Portuguese bought fumes, ivory, gems after the arduous and thence much

This trade gave merchants in Seville silver leaving America trade with Spain pesos passed around came from Peru time.) Various routes and Manila, ca ships might not less, might trade Finally, in a decade across the Pacific in 1631 banished had much effect was still allowed and silver from proved, on investment of silver that the

The great substantial means Much of the profits of the imports among themselves degree set the producers in America whether food, some estate owners grow more as the market structure provided in all

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instance, was 150 percent of the Spanish American valuation (8:1 v. 12:1).⁵² A broad range of goods bought in the Orient were amazingly cheap to those buying them with silver. And since the largest source of silver at the time was Peru, where Potosí was precisely in the 1570s experiencing its most spectacular boom of all time, the Mexico City merchants who started the Manila trade soon found themselves acting not so much as importers into New Spain as middle men in the exchange between Peru and Asia. Peruvian silver, transhipped into the galleons at Acapulco, spread from Manila throughout eastern Asia. Some of it was carried by Portuguese merchants working from the base at Macao that they had set up on the south China coast in the mid 1550s. The Portuguese bought silks at Canton nearby. These, together with porcelain, perfumes, ivory, gems, spices, and even iron and copper, arrived back in Acapulco after the arduous four- to seven-month arching traverse of the north Pacific; and thence much passed southward to Peru.

This trade grew with startling speed; indeed, for the authorities and the merchants in Seville, with thoroughly alarming speed. In 1597 the amount of silver leaving Acapulco for Manila was greater than the value of New Spain's trade with Spain. In 1602 the town council of Mexico City judged that 5 million pesos passed annually through Acapulco to the Philippines, 3 million of which came from Peru. (Potosí's production was some 6.9 million pesos a year at the time.) Various restrictions were tried: only two ships might ply between Acapulco and Manila, carrying westward a maximum of 500,000 pesos (1593); the two ships might not be larger than 200 tons (1604); only three ships, of 300 tons or less, might trade yearly between Peru and Mexico, and carry no silver (1604). Finally, in a desperate effort to stop the hemorrhage of South American silver across the Pacific, and at the insistence of the Seville merchant guild, the crown in 1631 banned trade between Peru and New Spain altogether. None of this had much effect, so powerful was the lure of profit in exporting silver. Trade was still allowed between Peru and Central America; transshipment of goods and silver from there to Mexico was simple.⁵³ The Manila galleon of 1646 proved, on investigation, to be carrying from six to fourteen times the amounts of silver that the consigning merchants had registered for export.⁵⁴

The great sums committed to the Philippine trade show how much silver the substantial merchants of Mexico City, and Lima too, had at their disposal. Much of the profit of the colonies' productive activities came to rest in the coffers of the import-export merchants, above all because they, collaborating among themselves and also with exporters in Seville, could to a considerable degree set the price of goods from Europe, as well as those from Asia. No producers in America had a comparable power to fix the price of their wares, whether food, cloth, craft items, or anything else. They certainly tried to do so; some estate owners, for instance, sought to extend their lands not so much to grow more as to prevent others from raising competing crops. But generally the market structures of the colonies, small and inefficient as they mostly were, provided in all but the short term some regulation of prices.

The profitability of the import business was clear from the earliest years. In 1529 the *alcalde mayor* of Oaxaca, and *encomendero* to boot, Juan Peláez de

Berrio, wrote to Spain: "It seems to me that one of the good and most important businesses in this land is merchandise, for in that are the true mines, beyond what one can believe ... The profit is so sure and so large that a well stocked shop here is the richest thing in the world, alchemy itself ... Would to God I had brought a thousand ducats' worth of clothing, for, as things are here, I could easily send back four, or even five, thousand from it, such is the need of Castilian clothes."⁵⁵ With time, local production expanded to cover everyday needs such as clothes; but even then importing merchants could still charge premium prices for luxuries not made in the Indies (or, even if made there, lacking the European or Oriental cachet): silks, taffeta, satin, velvet, damask, linen from Holland, Rouen cloth, Spanish paper, Spanish wine, decorative trimmings of silk, silver, and gold, kid gloves, hose, laced buskins, saffron, spices⁵⁶ – among much else.

INVESTMENT

Their profits made merchants a leading source of credit in the colonies. They lent officially at the legally approved annual rate of 5 percent, but in reality often at far higher interest if the risk seemed greater, to investors in property, mining, and all other lines of production. While this credit-supplying role of merchants is broadly clear, however, precisely how they performed it in the sixteenth century is far less so; and indeed the general topic of investment and capital formation in the early colonies is the least known, because least visible, aspect of their economic history.

It is, though, clear enough that most investment in the Spanish Indies in the sixteenth century was generated in America. The main exception was the period of Caribbean settlement, when funds came from Castilian, Italian, German, and Flemish sources.⁵⁷ Even then, however, gold-mining and agriculture could provide sizable accumulations, as Pizarro's funding of the Peruvian explorations and campaign from his landholdings in Panama shows. Loot from the mainland conquests then gave the lucky and the careful an opportunity to gather large quantities of gold, though their full potential as investment funds depended on their being hoarded until the extreme inflation of prices (enormous amounts of gold pursuing rare European goods) of the immediate conquest had passed. Once *encomienda* was in place on the mainland, Indian labor became the major source in the creation of fixed capital. Houses, storage buildings, irrigation works, early smelting apparatus, local roads – all these could be built with *encomienda* labor with little or no cash outlay by the *encomendero*. Canny *encomenderos* could also amass liquid capital in the form of bullion by sending their Indians to mine gold or silver, or by selling, for cash, in local markets other tribute items, such as pieces of cloth, or food, that Indians had paid in kind. Evidently enough, the growth of mining was central to any gathering of funds for investment, with silver even in its unminted form becoming (from the 1530s in New Spain and from the 1540s in Peru) a convenient means of accumulating value as well as of enabling exchange. Coined silver, of more

reliable value than the increasingly common ... Another mint was set up for the operations to Potosí, where

Mining seems to have been stimulated given it by the enormous content of the surface of the earth; therefore, was their enormous mineral with a very high value in reports. Initial profits of the workers received a large share when amalgamation by the other apparatus was later installed in places that proved too poor for smelting with mercury. Thus the while they were building, refined back into the position in Potosí, 1571–1572 42 percent of their after using amalgamation and the crown in the form of the treasury, the sole legal account until well into the free loans quite conscious

Rising silver output stimulated trade. In isolated regions of the empire, by the ways of the empire, by the medium of exchange, speaking, the second half of the Spanish Indies. For long-distance trade, exchange in Lima soon after 1600 500,000, and very many hoards, becoming one lender by that time with clergy. They, in the second yet fat, on gifts and bequests became a standard practice priests were, then, the Indies. No true banks large merchant houses in functions.

Wealthy merchants making small loans to people and cities across the empire

reliable value than the motley bits of metal that first served as money, became increasingly common with the foundation of a mint in Mexico City in 1536. Another mint was set up in Lima in 1565; but Viceroy Toledo moved its operations to Potosí, where striking of coin began in 1574.

Mining seems to have produced its own start-up capital (in addition to the stimulus given it by *encomienda* labor). Weathering had often raised the silver content of the surface ores that were usually the first to be worked. Not only, therefore, was their extraction cost minimal, but the early smelting was of mineral with a very high silver content, in places up to 50 percent by some reports. Initial profits could therefore be high, even if, as in the Andes, native workers received a large share of the product. Something similar happened when amalgamation began. Here the capital cost in milling machines and other apparatus was large. Typically, however, amalgamation processing was installed in places that possessed accumulations of already-mined ore that had proved too poor for smelting. Much of this material could be worked profitably with mercury. Thus the early amalgamators were spared the cost of extraction while they were building their refining mills, and plowed much of what they refined back into the producing apparatus. In the first five years of amalgamation in Potosí, 1571-5, records indicate that the miners reinvested some 42 percent of their after-tax production in building refining plant. Refiners using amalgamation also received help and, in effect, encouragement from the crown in the form of credit on the mercury they bought from the royal treasury, the sole legal supplier. The treasury tolerated large debts on this account until well into the next century. Those debts were, in reality, interest-free loans quite consciously made by the crown to boost silver production.

Rising silver output after the mid sixteenth century generally stimulated trade. In isolated regions far from mining centers and off the economic highways of the empire, barter persisted, and also the use of some non-metallic medium of exchange, such as cacao beans in southern Mexico. But, broadly speaking, the second half of the century brought a monetization, though uneven, of the Spanish Indies. From growing commerce, merchants, especially those in long-distance trade, extracted handsome profits. By one report there were in Lima soon after 1600 some traders with million-peso fortunes, many with 500,000, and very many with 100,000.⁵⁸ Merchants made loans from these hoards, becoming one major group of creditors in the colonies. The other large lender by that time was the church, and particularly the orders of regular clergy. They, in the second half of the sixteenth century, had grown stout, if not yet fat, on gifts and bequests in silver or property. Lending from these resources became a standard practice until the mid nineteenth century. Merchants and priests were, then, the closest approximation to bankers to emerge in the Indies. No true banks ever appeared in colonial times, though some of the large merchant houses in eighteenth-century Mexico performed many banking functions.

Wealthy merchants and the church were generally not in the business of making small loans to poor people. But local credit systems developed in towns and cities across the empire that did just that. The lenders were, again, typically

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merchants, but at the level of keepers of general stores (rather than of emporia selling expensive imported goods). They might sell on credit, or make small loans against the security of goods left by the borrowers; they were, therefore, acting as pawnbrokers. Particularly notable is the fact that many such lenders were women. In Potosí, where this petty-credit system has been closely studied, a good number of these women were Indians.⁵⁹ Their clients included both other Indians and Spaniards. Thus, before the sixteenth century ended, at the level of common, everyday existence there existed a remarkable integration in economic affairs between native people and colonizers. The case of Potosí also shows how quickly and easily native people could learn and adopt the new practices of monetized exchange and credit that the Spanish had introduced.

DOMINATION SURVEYED

Identification of resources, introduction of alien crops and animals, application of new productive tools and techniques, organization of labor, the mapping and plying of trade routes on land and sea, the opening of economic circuits through which flowed increasing charges of silver coin: all constituted a striking economic domination, even an economic re-formation, of Middle and South America in the sixteenth century. Economic matters were naturally linked with every other aspect of Spanish treatment of America and its native peoples in that first century. The church, while some of its members strove to protect Indians from labor exactions and the maltreatment that these often entailed, served broadly to bring native peoples into the Spanish physical and mental orbit. A parish priest might criticize *encomenderos* or the *mita*; but, at the same time, the more his efforts to protect Indians attached them to him, the more powerful an agent of Spanish presence he became. The state, too, was naturally and inextricably bound up with economic affairs. America's wealth, once identified in present and potential forms, became a central part of Spain's fiscal being. The government logically worked to raise its income from the Indies, as it gained the administrative means to do so, by modifying systems of production (in suppressing the *encomienda*, for example), maximizing the use of a dwindling labor supply (through setting up drafts of native workers), and adjusting taxes (lowering the silver royalty tax from a fifth to a tenth in Mexico, for instance, since the silver industry there, though productive, certainly did not boom on the scale of Potosí in the central Andes).

But Spanish domination was not by any means a simple imposition on America of alien methods and solutions. It was equally an adjustment and adaptation of European ways to New World conditions that sometimes came close to new invention. The American *audiencia* by the late sixteenth century had moved far from its Iberian source in becoming as much an executive as a juridical body. The estate structure of Spanish society shifted across the Atlantic, but the occupants of the estates changed, with ethnicity rather than function now becoming the determinant in placement. Spaniards' partiality for wheat resulted in the location of excellent wheat lands in New Spain and

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elsewhere; but colonists quite soon ate maize (and potatoes and manioc too) as well. The interesting, but unexploited, qualities of mercury long known in Europe were developed, in mid-sixteenth-century Mexico, into the basis of a silver-refining industry whose product had worldwide economic effects. Even long-accepted European notions of what constituted humanity necessarily expanded when Spaniards met people whose qualities contradicted the old patterns. In sum, the same capacity to learn, to improvise, to respond creatively to the unknown and the uncertain that had given the conquistadores a crucial edge over the native states persisted among early residents of the Indies, both officials and settlers, advancing the processes of domination that are the essence of Spanish American history between the conquests and the end of the sixteenth century.

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